MILITARY SPECIFICATION

FUZE, POINT DETONATING, M739A1 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the M739A1 Point Detonating Artillery Ammunition Fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-60412	-Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes, General Specification for

STANDARDS

MILITARY

MIL-STD-331	-Fuze and Fuze Components, Environmental and Performance Tests for

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

ARRADCOM

7548099	-Box, Packing, Ammunition
7548103	-Box, Set-Up, Packing, Ammunition, Outer
7548104	-Box, Set-Up, Packing, Ammunition, Inner
9255165	-Lead Assembly
9345332	-Fuze, Point Detonating, M739A1

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material and parts. Material and parts shall be in accordance with applicable drawings, standards and specifications.

3.2 Fuze assembly. The M739A1 Point Detonating fuze assemblies shall comply with all requirements specified on drawing 9345332 and with all requirements specified in applicable specifications and standards.

3.3 Transportation vibration. The fuze assembly shall comply with the requirements of MIL-STD-331, Test 104, Procedure II.

3.4 Proving ground.

3.4.1 Premature burst. There shall be no premature burst in the gun bore or in the flight before impact.

3.4.2 Functioning.

3.4.2.1 Superquick. The fuzes shall cause the projectile to function high order and shall show no evidence of reverse functioning (see 6.4).

3.4.2.2 Delay. The fuze shall cause the projectile to function high order.

3.4.3 Non-functioning. The fuze shall not function upon impact with the target when subjected to this test.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:03Z
Check the source to verify that this is the currentversion before use.

3.5 Lead assembly output test. The lead assembly shall function high order as evidenced by a minimum .210 inch diameter hole in a one-eighth-inch-thick lead disc.

3.6 Workmanship. All parts and assemblies shall be loaded and assembled in a thorough, workmanlike manner and all manufacturing, processing and assembling operations shall be correctly performed. All parts shall be free of burrs, sharp edges, cracks, dirt, grease, dust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 (PA) shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following types of inspections shall be conducted on this item:

a. Quality Conformance Inspection

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 (PA). In addition, each lot of M739A1 fuzes shall contain:

a. Fuze components of one lot interfix number from one manufacturer.

b. Booster pellets of one lot interfix number from one manufacturer.

4.3.2. Examination. See MIL-A-48078 (PA).

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.1	Pellet, Booster		9255168
			NEXT HIGHER ASSEMBLY 9345332

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cracked, chipped or broken		0.40%	3.2	Visual
102	Weight		0.40%	3.2	Balance
103	Height		0.40%	3.2	Gage
104	Diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63519 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9345331
4.3.2.2	Fuze Body Assembly				NEXT HIGHER ASSEMBLY 9345332
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
102	External fuze body threads damaged		0.40%	3.2	Visual
131	Internal fuze body threads damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:03Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.3	Fuze, PD, M739A1, Prior to Assembling Booster Cup				9345332
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Booster pellet improperly oriented in booster cup		0.40%	3.2	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-F-63519 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9345332
4.3.2.4	Fuze, PD, M739A1				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
131	Length from base of flange to base of booster cup		0.40%	3.2	Gage
132	Sealant not 360° around booster cup		0.40%	3.2	Visual
Minor					
201	Marking missing or misleading		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9345332
4.3.2.5	Fuze, PD, M739A1 Destructive Lot Acceptance Test				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
2	Transportation Vibration Test, MIL-STD-331, Test No. 104, Procedure II	10		3.3	4.4.2
3	Premature Burst	35 (a)		3.4.1	4.4.4.2
4	Non-Functioning Test	10		3.4.3	4.4.4.2
Major					
101	Functioning-SQ	25 (b)		3.4.2.1	4.4.4.2
102	Reverse Functioning	25 (b)		3.4.2.1	4.4.4.2
104	Transportation Vibration Test, MIL-STD-331, Test No. 104, Procedure II	10		3.3	4.4.2/4.4.4.2

NOTES: (a) Premature burst information is to be taken when firing tests 4 and 101.
(b) Ten of these fuzes are from test 104.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63519 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.2.6	Cup, Lead		9255166
			NEXT HIGHER ASSEMBLY 9255165

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of bottom of cup		0.40%	3.1	Gage
131	Length		0.40%	3.1	Gage
132	Inside diameter		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-25T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9255167
4.3.2.7	Disc, Lead				NEXT HIGHER ASSEMBLY 9255165
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
131	Thickness		0.40%	3.1	Gage
132	Diameter, minimum		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-F-63519 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9255165
4.3.2.8	Lead Explosive PA508				NEXT HIGHER ASSEMBLY 9294833
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cracks or splits		100%	3.2	Visual
Major					
101	Weight of Composition A5		0.40%	3.2	Balance
103	Identification missing or color incorrect		0.40%	3.2	Visual
104	Assembly distorted or damaged to the extent that functioning may be improper		0.40%	3.2	Visual
131	Diameter, maximum		0.40%	3.2	Gage
132	Overall length, maximum		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63519 (AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER 9255165
4.3.2.8.1	Lead Explosive PA 508 (Prior to Disc Ass'y.)				NEXT HIGHER ASSEMBLY 9255165
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Density of explosive in lead cup	5		3.2	4.4.8

NOTES:
Sample production at rate of 5 every 4 hours as described in paragraph 4.4.8. This paragraph applies only when alternate method of loading is used (see Dwg. 9255165).

DT MRB-QA Form 100 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which Is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63519 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.9	Lead Assembly, Destructive Lot Acceptance Test				9255165
					NEXT HIGHER ASSEMBLY 9294833
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Output test	150		3.5	4.4.5

NOTES:

DLA-CA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 which is Obsolete

Source: https://assist.dla.mil - Downloaded: 2026-03-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548104
4.3.2.10	Box, Set-up, Packing, Ammunition, Inner				NEXT HIGHER ASSEMBLY 7548103
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
131	Box damaged exposing contents		0.40%	3.1	Visual
132	Label missing or incorrect		0.40%	3.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.11	Box, Set-up, Marking Ammunition, Outer	1 OF 1	7548103
			NEXT HIGHER ASSEMBLY 7548099

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
131	Barrier bag damaged		0.40%	3.1	Visual
132	Label missing or incorrect		0.40%	3.1	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63519 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.12	Box, Packing, Ammunition	1 OF 1	7548099
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
131	Box damaged		0.40%	3.1	Visual
132	Marking inadequate		0.40%	3.1	Visual
133	Strapping missing or loose		0.40%	3.1	Visual
134	Marking incomplete, misleading or unidentifiable		0.40%	3.1	Visual

NOTES:

Source: https://assist.dla.mil/ -- Downloaded: 2020-10-23T03:05Z
Check for source(s) to verify that this is the current version before use.

4.3.3 Testing.

4.3.3.1 Transportation vibration.

4.3.3.1.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have been found acceptable, a sample of fifteen (15) loaded fuzes shall be randomly selected from each lot for the test. If any assembly fails to comply with the requirements of the test, the lot shall be rejected.

4.3.3.1.2 After three (3) consecutive lots have complied with the criteria of 4.3.3.1.1, the sample size shall be as specified in 4.3.2.5. If any assembly fails to comply with the requirements of the test, the lot shall be rejected and testing shall revert to 4.3.3.1.1.

4.3.3.2 Proving Ground tests.

4.3.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have been found acceptable, a sample size of seventy (70) shall be randomly selected from each lot for the test. The lot shall be rejected if the requirements of 4.4.4.1 are not met.

4.3.3.2.2 After three (3) consecutive lots have complied with the criteria specified in 4.3.3.2.1, the sample size shall be as specified in 4.3.2.5. The lot shall be rejected if the requirements of 4.4.4.2 are not met and testing shall revert to 4.3.3.2.1.

4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Investigation Method" column in the tables starting with paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 (PA) and 6.3 herein.

4.4 Test methods and procedures.

4.4.1 Intentionally omitted.

4.4.2 Transportation vibration test. The loaded fuze assemblies shall be subjected to the transportation vibration test in accordance with MIL-STD-331, Test 104, Procedure II. The sample shall then be radiographically inspected in two planes as follows to determine compliance with the safety aspects of the test:

a. Horizontal Plane - Screwdriver slot facing viewer.
b. Horizontal Plane - Screwdriver slot at left.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:03Z
Check the source to verify that this is the current version before use.

As a minimum inspection requirement for Critical Defects, no S&A shall be in an armed or partially armed condition. As a minimum inspection requirement for Major Defects, the x-ray shall indicate that no metal parts components are broken. Fuzes subject to this test shall be forwarded to the Proving Ground for testing as specified in 4.4.4.

4.4.3 Intentionally omitted.

4.4.4 Proving Ground tests.

4.4.4.1 The fuzes shall be assembled to the appropriate HE loaded projectiles, except for the non-functioning test where inert loaded projectiles with T2 supplementary charges shall be used, and fired from the weapons specified below. Target distances for all phases except for the one (1) inch plywood screen shall be any convenient distance. The quadrant elevations shall be as specified below. For all ground impact functioning, appropriate geophonic sensing devices shall be employed to verify high order functioning of the projectile upon impact. The acceptance criteria is as follows:

DEFECT	ACCEPT	REJECT
Premature Burst	0	1
Non-Functioning	0	1
Duds (see 6.6)	1	2
Reverse Functioning	1	2

WEAPON	ZONE	QUADRANT ELEVATION	TARGET	FUZE SETTING	SAMPLE SIZE
Non Functioning:					
105MM, M102	7		1" Plywood @ 115 Feet	SQ	10
Functioning:					
105MM, M102	7	Low	Ground	SQ	20*
105MM, M102	7		1" Plywood @ 500 Feet	SQ	10
155MM, M109A1	2	Low	Ground	SQ	20**

* 10 Transportation Vibration Conditioned Fuzes
** 5 Transportation Vibration Conditioned Fuzes

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

4.4.4.2 The testing shall be conducted as specified in 4.4.4.1 except that the acceptance criteria and weapon firings shall be as specified below.

DEFECT	ACCEPT	REJECT
Premature Burst	0	1
Non-Functioning	0	1
Duds (see 6.6)	0	2**
Reverse Functioning	1	2

WEAPON	ZONE	QUADRANT ELEVATION	TARGET	FUZE SETTING	SAMPLE SIZE
Non Functioning:					
105MM, M102	7		1" Plywood @ 115 Feet	SQ	10
Functioning:					
105MM, M102	7	Low	Ground	SQ	10*
105MM, M102	7		1" Plywood @ 500 Feet	SQ	5
155MM, M109A1	2	Low	Ground	SQ	10

* 10 Transportation Vibration Conditioned Fuzes
** If only one dud occurs during the Functioning tests, repeat the Functioning tests with a second sample of 25 fuzes. If no additional duds occur, accept the lot.

If any production lot fails to meet the above criteria, testing shall revert to 4.4.4.1 until three (3) consecutive lots have been accepted.

4.4.5 Lead assembly output test. The lead assembly output test shall be performed using the appropriate test equipment and a one-eighth-inch-thick lead disc. High order functioning shall be evidenced by a minimum .210 inch diameter hole in the lead disc. If two or more lead assemblies fail to meet the minimum diameter hole requirement, the lot shall be rejected.

4.4.6 Intentionally omitted.

4.4.7 Intentionally omitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

4.4.8 Density of explosive in lead cup (when loaded by alternate method). A minimum of five (5) assemblies, representative of the explosive leads being processed, shall be loaded and the densities determined to qualify each press prior to its use in production. In addition, a sample of five (5) leads shall be processed during the last portion of each four (4) hour's production from each press and density determined as a control of each press during production.

If any defective lead is found when inspection is performed for control of a press, the press producing the defective leads shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All presses removed from production because of some fault, may after corrective action be returned to production providing they pass the qualification test above.

The samples shall be selected prior to assembly of lead disc and crimping and the following procedure used to determine the density of the explosive charge.

1. Measure and record height (h) from base of cup to top of explosive column to .001 cm.

2. Weigh and record weight of loaded cup to 0.001 gram.

3. Remove explosive from cup.

4. Weight empty cup to 0.001 gram.

5. Measure inside diameter (d) of empty cup to .001 cm.

6. Measure end thickness of empty cup to .001 cm.

7. Calculate weight of explosive by subtracting the value obtained in (4) above from the value obtained in (2) above.

8. Calculate the explosive column height by subcontracting the cup thickness measured in (6) above from the total height (h) measured in (1) above.

9. Calculate the explosive volume using the formula $V = \pi \cdot r^2(h)$ (h is the value obtained in (8) above and r = d/2 from (5) above.

10. Calculate the density by dividing the weight obtained in (7) above by the volume obtained in (9) above.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:03Z
Check the source to verify that this is the currentversion before use.

If the density of any explosive lead falls outside the density limits specified on the applicable drawing, the lead shall be classed defective. The contractor may use another method for determining density provided approval of the method has been obtained from the contracting officer.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. The fuzes shall be preserved and packaged in accordance with the requirements and inspection procedures specified in MIL-P-60412 for Fuze, PD, M739.

5.1.2 Level C. The lead assemblies shall be preserved and packaged in accordance with drawings 7548103 and 7548104.

5.2 Packing.

5.2.1 Level A. The fuzes shall be packed in accordance with the requirements and inspection procedures specified in MIL-P-60412 for Fuze, PD, M739.

5.2.2 Level C. The lead assemblies shall be packed in accordance with drawing 7548099.

5.3 Marking.

5.3.1 Level A. The fuze packaging and packing shall be marked in accordance with the requirements and inspection procedures specified in MIL-P-60412 for Fuze, PD, M739.

5.3.2 Level C. The lead assembly packaging and packing shall be marked in accordance with drawings 7548099, 7548103 and 7548104.

5.4 General.

5.4.1 Exterior containers. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock-numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with MIL-P-60412 and as specified herein.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on Artillery Ammunition.

Source: https://assist.dla.mil - Downloaded: 202640-23T03:03Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. See MIL-A-48078 (PA).

6.3 Submission of designs for approval. See MIL-A-48078 (PA). Submit equipment designs, as required, to: Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Reverse functioning. Reverse functioning is defined as "Delay" functioning of the fuze when set in the Superquick position.

6.5 Proving ground test summary.

QUALITY CONFORMANCE TEST

FIRST 3 CONSECUTIVE ACCEPTABLE LOTS

ACCEPTANCE CRITERIA

DEFECT	ACCEPT	REJECT
Premature Burst	0	1
Non-Functioning	0	1
Duds (see 6.6)	1	2
Reverse Functioning	1	2

TESTING

WEAPON	ZONE	QUADRANT ELEVATION	TARGET	FUZE SETTING	SAMPLE SIZE
Non Functioning:					
105MM, M102	7		1" Plywood @ 115 Feet	SQ	10
Functioning:					
105MM, M102	7	Low	Ground	SQ	20*
105MM, M102	7		1" Plywood @ 500 Feet	SQ	10
155MM, M109A1	2	Low	Ground	SQ	20**

* 10 Transportation Vibration Conditioned Fuzes
** 5 Transportation Vibration Conditioned Fuzes

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

SUBSEQUENT LOTS

ACCEPTANCE CRITERIA

DEFECT	ACCEPT	REJECT
Premature Burst	0	1
Non-Functioning	0	1
Duds (see 6.6)	0	2**
Reverse Functioning	1	2

TESTING

WEAPON	ZONE	QUADRANT ELEVATION	TARGET	FUZE SETTING	SAMPLE SIZE
Non Functioning:					
105MM, M102	7		1" Plywood @ 115 Feet	SQ	10
Functioning:					
105MM, M102	7	Low	Ground	SQ	10*
105MM, M102	7		1" Plywood @ 500 Feet	SQ	5
155MM, M109A1	2	Low	Ground	SQ	10

* 10 Transportation Vibration Conditioned Fuzes
** If only one dud occurs during the Functioning tests, repeat the Functioning tests with a second sample of 25 fuzes. If no additional duds occur, accept the lot.

6.6 Definition of dud. Fuzes which fail to function where intended shall be classified as duds. For Superquick (SQ) firings into a plywood target at 500 feet, intended functioning of the fuze is on impact with the target, i.e., on the target.

Custodian:
Army-AR

Preparing activity:
Army-AR

Project: 1390-A310

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:03Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
US ARMY ARMAMENT RESEARCH & DEVELOPMENT COMMAND
ATTN: DRDAR-QAR
DOVER, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-63519 (AR) Fuze, Point Detonating, M739A1 Loading, Assembling and Packing

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:03Z
Check the source to verify that this is the current version before use.